IoT-Based Smart Obstacle Avoiding Robot Using Arduino UNO and ESP32 for Real-Time Remote Monitoring

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Abstract

The autonomous small-scale robot may cost too much, be too difficult to make and lack any remote realtime control. The presented work offers the IoT based smart obstacle avoiding robot, which uses inexpensive materials. For the robot movement control the local microcontroller Arduino UNO board with the Adafruit Motor Shield V1 (L293D) was chosen and four TT geared DC motors are used. The robot is equipped by HC-SR04 ultrasonic sensor with SG90 servo mounted on it for measuring the distance ahead. When the obstacle appears near enough within the 30 cm safe distance the robot stops and moves backwards slightly to check left and right obstacles and then moves towards free paths. The ESP32 Wi-Fi module takes the data from the UNO concerning the distance and status of the robot and transfers it to the ThingSpeak cloud dashboard to provide remote monitoring of the robot status. There are two independent 6 V and 5 V buck voltage converters for motors and logic boards correspondingly to increase power stability. The robot is reactive, lightweight and suitable for educational and inspection route at the warehouse aisles, but it is not a safety critical industrial vehicle.

Keywords: Arduino UNO; ESP32; Internet of Things (IoT); Obstacle Avoiding Robot; HC-SR04 Ultrasonic Sensor; L293D Motor Driver; ThingSpeak

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